

Four interpretations of Energinet’s P90 reliability requirement for FCR-D reserve bidding

Abstract

The P90 criterion underlying Danish/Nordic FCR-D prequalification is conventionally summarised as “the reserved capacity must be available at least 90 % of the time.” When the available flexibility is a stochastic process indexed by minute and by realisation of the underlying fleet uncertainty, this verbal requirement admits at least four mathematically distinct encodings. We state the four formulations as chance constraints, give their scenario-based mixed-integer reformulations, and use four worked examples to show that the four are partially but not totally ordered: the joint chance constraint and the per-scenario time-availability constraint are formally incomparable, and the average/sum constraint sits strictly below all of the others. The choice of formulation therefore changes both the admissible bid level and the physical guarantee the bidder is delivering.

1 Energinet’s P90 requirement: “90 % reliability”

In the Danish/Nordic FCR-D market, prequalified reserve providers must deliver their bid capacity reliably during activation. The P90 prequalification criterion is conventionally summarised as

the reserved capacity must be available at least 90 % of the time.

For a deterministic resource on a deterministic time grid this would be unambiguous. For a Limited Energy Reservoir (LER) such as an EV aggregator, however, the available flexibility F_m at minute m within the delivery window $\{1, \dots, M\}$ is a random variable: it depends on the stochastic state of charge, plug-in pattern, and consumption profile of the fleet. “Available 90 % of the time” then admits at least the four readings collected in Table 1.

<i>Plain-English reading of “available 90 % of the time”</i>	
(1)	Within <i>every</i> realisation of the fleet, the bid is deliverable in at least 90 % of the minutes of the delivery window.
(2)	With probability at least 0.90 over the fleet uncertainty, the bid is deliverable in <i>every</i> minute of the delivery window.
(3)	For <i>each minute taken separately</i> , the bid is deliverable with probability at least 0.90.
(4)	Averaged over minutes and realisations, the bid is deliverable at least 90 % of the time.

Table 1: Four English interpretations of the P90 criterion. They sound similar but enforce inequivalent things.

The four readings each correspond to a different chance-constraint formulation. They yield different feasible regions in the bid variable, different optimal bid levels, and different physical guarantees once the bid is committed. The remainder of this note states the four formulations precisely, gives their scenario-based reformulations, and shows via four illustrative cases that no total order on their strictness exists.

2 The four chance-constraint formulations

Setup. Let $c \geq 0$ denote the reserve capacity bid (a scalar decision variable). Let $F_1, \dots, F_M$ be the random available flexibility over the M minutes of the delivery window, distributed according to a joint law $\mathcal{D}$. Let $\varepsilon = 0.10$ be the violation tolerance corresponding to “P90.”

For the scenario-based reformulations, Ω scenarios $\{F_m^{(\omega)}\}_{m=1, \dots, M; \omega=1, \dots, \Omega}$ are drawn i.i.d. from $\mathcal{D}$, the binary $y_{m,\omega} \in \{0, 1\}$ flags the violation event $c > F_m^{(\omega)}$, and N is a sufficiently large constant for the big- M implication.

The four formulations are presented in numerical order, from the most operationally direct reading of the P90 wording to the most permissive.

Formulation (1): per-scenario time availability

Chance-constraint statement.

$$\frac{1}{M} \sum_{m=1}^M \mathbf{1}[c \leq F_m] \geq 1 - \varepsilon \quad \text{almost surely.} \quad (1)$$

Interpretation. In every realisation ω of the fleet uncertainty, the fraction of minutes in which the bid is deliverable must be at least $1 - \varepsilon$. There is no probabilistic slack across realisations: each ω must individually clear the 90% time-availability bar. This is the strictest interpretation of “of the time” read as a frequency *within* each delivery window.

Scenario reformulation.

$$c - F_m^{(\omega)} \leq y_{m,\omega} N, \quad \forall m, \omega, \quad (2)$$

$$\sum_{m=1}^M y_{m,\omega} \leq \varepsilon M, \quad \forall \omega. \quad (3)$$

The budget on bad minutes is enforced *inside every scenario*, with no scenario-level slack.

Formulation (2): joint chance constraint

Chance-constraint statement.

$$\mathbb{P}(c \leq F_m \ \forall m = 1, \dots, M) \geq 1 - \varepsilon. \quad (4)$$

Interpretation. With probability at least $1 - \varepsilon$, the bid is deliverable in *every single* minute of the hour. The probability is over realisations ω , and within each “good” realisation the constraint must hold for all m simultaneously. A fraction up to ε of realisations may fail entirely; the rest must be flawless throughout.

Scenario reformulation. Introduce a single binary $z_\omega \in \{0, 1\}$ per scenario, shared across m :

$$c - F_m^{(\omega)} \leq z_\omega N, \quad \forall m, \omega, \quad (5)$$

$$\sum_{\omega=1}^{\Omega} z_\omega \leq \varepsilon \Omega. \quad (6)$$

A scenario is consumed from the budget the moment any one of its minutes fails. The fact that z_ω is shared across m is essential: it couples the constraints, encoding the joint event “some m violates” rather than individual events. Replacing z_ω by $y_{m,\omega}$ would weaken (4) to formulation (3).

Formulation (3): individual (per-minute) chance constraints

Chance-constraint statement.

$$\mathbb{P}(c \leq F_m) \geq 1 - \varepsilon, \quad \forall m = 1, \dots, M. \quad (7)$$

Interpretation. For each minute taken *separately*, the bid is delivered with probability at least $1 - \varepsilon$. The constraints decouple across m : the realisations that fail in minute m may differ entirely from those that fail in minute m' . Consequently, the probability that the bid is delivered across *all* minutes simultaneously can drop as low as $1 - M\varepsilon$ (Bonferroni bound), much weaker than the $1 - \varepsilon$ guarantee of (2).

Scenario reformulation.

$$c - F_m^{(\omega)} \leq y_{m,\omega} N, \quad \forall m, \omega, \quad (8)$$

$$\sum_{\omega=1}^{\Omega} y_{m,\omega} \leq \varepsilon \Omega, \quad \forall m. \quad (9)$$

A separate budget per minute, with no coupling across m .

Formulation (4): sum / expected-violation constraint

Chance-constraint statement.

$$\frac{1}{M} \sum_{m=1}^M \mathbb{P}(c \leq F_m) \geq 1 - \varepsilon \iff \mathbb{E} \left[\sum_{m=1}^M \mathbf{1}[c > F_m] \right] \leq \varepsilon M. \quad (10)$$

Interpretation. The *average* per-minute reliability is at least $1 - \varepsilon$, equivalently the expected number of violated minutes per delivery window is at most εM . Strictly speaking this is not a chance constraint but a constraint on an expectation. Failures may concentrate on a single minute, on a few catastrophic realisations, or be spread thinly — only the total budget matters.

Scenario reformulation.

$$c - F_m^{(\omega)} \leq y_{m,\omega} N, \quad \forall m, \omega, \quad (11)$$

$$\sum_{m=1}^M \sum_{\omega=1}^{\Omega} y_{m,\omega} \leq \varepsilon M \Omega. \quad (12)$$

A single global budget on violations across all (m, ω) pairs.

3 Strength ordering

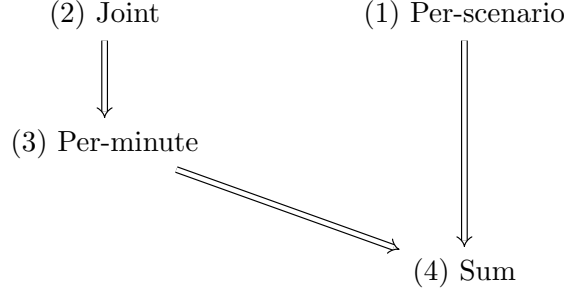
The four formulations are partially ordered by strictness, but *not* totally ordered. Throughout this section “ $A \Rightarrow B$ ” means: every c feasible for A is feasible for B (so A has the smaller feasible region and is the stricter condition).

3.1 Implications that hold

- (1) (2) $\Rightarrow$ (3). If at most $\varepsilon \Omega$ scenarios contain any violation, then in every minute the violations come from this set, so $\sum_{\omega} y_{m,\omega} \leq \varepsilon \Omega$ for every m .
- (2) (3) $\Rightarrow$ (4). Summing the per-minute bound across M minutes gives the total bound: $\sum_{m,\omega} y_{m,\omega} \leq M \cdot \varepsilon \Omega = \varepsilon M \Omega$.
- (3) (1) $\Rightarrow$ (4). Symmetrically, $\sum_{m,\omega} y_{m,\omega} = \sum_{\omega} (\sum_m y_{m,\omega}) \leq \Omega \cdot \varepsilon M = \varepsilon M \Omega$.

3.2 Hasse diagram

Collecting these implications gives the partial order shown below. Arrows point from stricter to weaker.



(4) is the unique weakest element. (3) sits strictly below (2). (1) sits strictly below (4) on a separate branch. Crucially (1) is *neither* above nor below (2) nor (3): the per-scenario constraint inhabits an arm of the partial order distinct from the chain $(2) \Rightarrow (3) \Rightarrow (4)$. The four illustrative cases below are minimal witnesses for the missing implications.

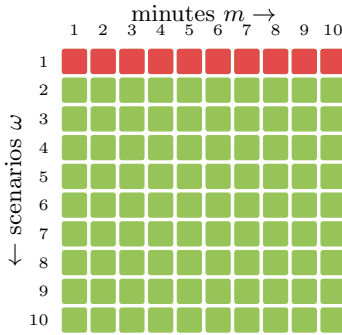
3.3 Four illustrative cases

We work with $M = \Omega = 10$ and $\varepsilon = 0.10$, so the budgets are

$$\varepsilon M = \varepsilon \Omega = 1, \quad \varepsilon M \Omega = 10.$$

Each cell (m, ω) in the heatmaps below is one realised pair:  $c \leq F_m^{(\omega)}$ (constraint satisfied),  $c > F_m^{(\omega)}$ (constraint violated). Rows index scenarios $\omega \in \{1, \dots, 10\}$ and columns index minutes $m \in \{1, \dots, 10\}$.

Case A: only the per-scenario constraint (1) fails. A single scenario fails in every minute; all other scenarios are perfect. The configuration is one fully-red row.

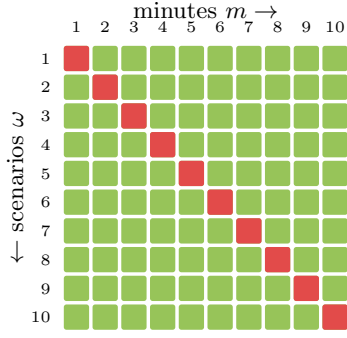


Formulation	Quantity	Verdict
(1) Per-scenario	$\max_{\omega} \sum_m y_{m,\omega} = 10 \not\leq 1$	violated
(2) Joint	$ \{\omega : \exists m, y_{m,\omega}=1\} = 1 \leq 1$	satisfied
(3) Per-minute	$\max_m \sum_{\omega} y_{m,\omega} = 1 \leq 1$	satisfied
(4) Sum	$\sum_{m,\omega} y_{m,\omega} = 10 \leq 10$	satisfied

Figure 1: Case A. One scenario (row 1) fails in every minute; all other scenarios are perfect.

The single catastrophic scenario consumes the entire joint budget exactly, which is enough to keep (2) satisfied. Each minute has only one failing scenario, so (3) is also at the boundary. The total is $10 \leq \varepsilon M \Omega$, so (4) passes too. But conditional on the unreliable scenario, no time-availability is delivered, and (1) is violated by a factor of ten. Case A is therefore a witness that $(2) \not\Rightarrow (1)$ and $(3) \not\Rightarrow (1)$.

Case B: only the joint chance constraint (2) fails. Every scenario has exactly one bad minute, and that bad minute is different for every scenario. The configuration is the diagonal of the matrix. Each scenario delivers in nine of ten minutes, so (1) is satisfied at the boundary; each minute fails in only one scenario, so (3) is at the boundary too; and the total of ten violations matches the sum



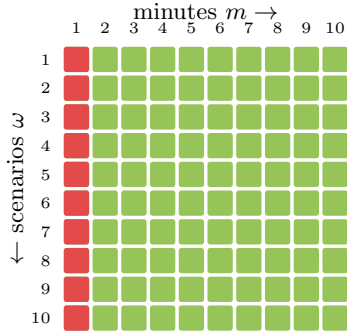
Formulation	Quantity	Verdict
(1) Per-scenario	$\max_{\omega} \sum_m y_{m,\omega} = 1 \leq 1$	satisfied
(2) Joint	$ \{\omega : \exists m, y_{m,\omega}=1\} = 10 \not\leq 1$	violated
(3) Per-minute	$\max_m \sum_{\omega} y_{m,\omega} = 1 \leq 1$	satisfied
(4) Sum	$\sum_{m,\omega} y_{m,\omega} = 10 \leq 10$	satisfied

Figure 2: Case B. Each scenario has exactly one bad minute, but every scenario fails at a different time.

budget exactly. But *every* scenario contains at least one violated minute, which is precisely the event the joint constraint forbids in more than $\varepsilon\Omega = 1$ of the scenarios. Case B is a witness that $(1) \not\Rightarrow (2)$ and $(3) \not\Rightarrow (2)$.

Cases A and B together prove that (1) and (2) are incomparable: A is feasible for (2) but not (1), B is feasible for (1) but not (2).

Case C: per-minute (3) constraints fail. Every scenario has exactly one bad minute, and that bad minute is the *same* for all scenarios. The configuration is one fully-red column.



Formulation	Quantity	Verdict
(1) Per-scenario	$\max_{\omega} \sum_m y_{m,\omega} = 1 \leq 1$	satisfied
(2) Joint	$ \{\omega : \exists m, y_{m,\omega}=1\} = 10 \not\leq 1$	violated
(3) Per-minute	$\max_m \sum_{\omega} y_{m,\omega} = 10 \not\leq 1$	violated
(4) Sum	$\sum_{m,\omega} y_{m,\omega} = 10 \leq 10$	satisfied

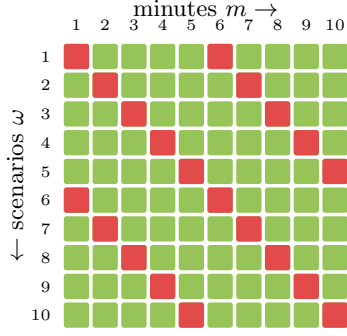
Figure 3: Case C. Every scenario fails at the same minute (column 1); all other minutes are perfect in every scenario.

Each scenario fails one minute, so (1) and (4) are at the boundary. But *all ten* scenarios fail at the same minute, which violates the per-minute budget by tenfold. The joint constraint (2) is consequently also violated: any configuration that fails (3) must fail (2), since $(2) \Rightarrow (3)$ makes (2) a stronger constraint. The reverse direction does not hold — Case B already showed that (2) failing does not force (3) to fail. Case C is also a witness that $(1) \not\Rightarrow (3)$: per-scenario availability is fully respected even though the per-minute constraint collapses.

Cases A and C together prove that (1) and (3) are incomparable: A is feasible for (3) but not (1), C is feasible for (1) but not (3).

Case D: the average/sum constraint (4) fails. Every scenario has two bad minutes, distributed so that every minute is bad in two scenarios. The total of twenty violations exceeds the sum budget of ten.

Once the sum constraint is violated, none of (1), (2), or (3) can hold either: (4) is the unique weakest element of the partial order, so any configuration that exceeds the sum budget must *a fortiori* exceed the per-scenario, joint, and per-minute budgets too. Case D illustrates this directly — all four constraints are violated by a factor of two or more — and confirms that the chain $(2) \Rightarrow (3) \Rightarrow (4)$ and



Formulation	Quantity	Verdict
(1) Per-scenario	$\max_{\omega} \sum_m y_{m,\omega} = 2 \not\leq 1$	violated
(2) Joint	$ \{\omega : \exists m, y_{m,\omega}=1\} = 10 \not\leq 1$	violated
(3) Per-minute	$\max_m \sum_{\omega} y_{m,\omega} = 2 \not\leq 1$	violated
(4) Sum	$\sum_{m,\omega} y_{m,\omega} = 20 \not\leq 10$	violated

Figure 4: Case D. Each scenario has two bad minutes, evenly distributed so that each minute is bad in two scenarios. Total: twenty violations.

the edge (1) $\Rightarrow$ (4) are tight: the moment (4) fails, everything strictly above (4) in the diamond fails as well.

4 Practical implication

The four formulations offer different trade-offs between conservativeness, computational tractability, and the type of guarantee delivered. Under the same dataset, they will in general yield four different admissible bid levels; the choice between them depends on the interpretation of P90 the bidder wishes to encode and on the operational trade-offs deemed acceptable.

Formulation (1) — per-scenario time availability. The most conservative model with respect to scenario uncertainty: every realisation must individually clear the 90 % within-window bar, with no probabilistic slack. The reformulation places one budget inside each scenario. Aligns with the reading “*in every operating hour the reserve must be delivered for at least 90 % of the minutes.*”

Formulation (2) — joint chance constraint. Less conservative than (1) across scenarios — a fraction up to ε of realisations may fail entirely — but equally conservative within the surviving realisations, which must be flawless throughout. The reformulation requires one binary per scenario shared across all minutes; this coupling makes it computationally more involved than the relaxations below. Aligns with the reading “*with probability at least 0.9 over fleet uncertainty, the reserve is deliverable throughout the hour.*”

Formulation (3) — individual (per-minute) chance constraints. A relaxation of (2): the per-minute constraints decouple, so different minutes may fail in different scenarios. The reformulation has M independent budgets, the most tractable structure among (1)–(3). The Bonferroni lower bound on joint reliability across the window, $1 - M\varepsilon$, loosens as M grows; under correlated scenarios the actual joint reliability may be tighter. Aligns with the reading “*for each minute taken separately, the bid is deliverable with probability at least 0.9.*”

Formulation (4) — sum / expected-violation constraint. A relaxation that caps the expected number of violated (m,ω) pairs rather than a probability or a frequency. The reformulation is a single linear budget — the most tractable form of the four — and failures may concentrate in particular minutes or scenarios as long as the global budget holds. Aligns with the reading “*averaged over minutes and realisations, the bid is deliverable at least 90 % of the time*” and is appropriate when the penalty for violations is linear in their count.

The four are not interchangeable: (1) and (2) encode two distinct strict interpretations of the P90 wording, and (3) and (4) encode two distinct relaxations that trade conservativeness for tractability or for a different definition of reliability. The bidder’s choice determines both the admissible bid level and the type of guarantee committed to.

5 Ex-post audit behaviour

A natural follow-up question is whether the four formulations would agree if the regulator audited a bidder over many delivery windows — say a month of operation — and computed the realised reliability. The answer depends on (a) which audit metric is computed and (b) whether the fleet uncertainty has structure across minutes and across hours.

Each formulation matches a natural audit metric. By the law of large numbers, a bid placed at the binding constraint of formulation (k) will, after many delivery windows, yield a violation rate close to ε *when the realised data is summarised by the audit metric naturally associated with (k)* :

Formulation	Natural audit metric over a month
(1) Per-scenario	worst-hour fraction of bad minutes (max over delivery windows)
(2) Joint	fraction of hours with at least one bad minute
(3) Per-minute	worst minute-slot’s violation rate (max over $m = 1, \dots, M$)
(4) Sum	overall average violation rate across all (hour, minute) pairs

When the four audits agree. If fleet uncertainty is exchangeable across minutes and across hours — no minute slot is structurally harder than another, no day type (weekday/weather/season) is systematically catastrophic, and violations are rare and roughly independent — the four metrics approximately coincide. In this regime any of the four formulations passes any of the four audits at roughly the ε level, and the choice of formulation is mostly a tractability decision.

When the four audits diverge. For real EV-aggregator data the assumptions above typically fail, and the audit number depends sharply on both the formulation that built the bid and the metric the regulator uses:

- *Time-of-day effects within the hour* (e.g. minute 1 just after activation is tighter than minute 8 due to plug-out timing). A (4)-bid allows the structurally hard minute to absorb most violations, so a per-minute audit may read a violation rate well above ε on the worst minute slot while the overall-average audit reads ε . A (3)-bid forces every minute slot toward ε .
- *Catastrophic-day effects* (cold mornings, holidays, atypical fleet states). A (2)-bid loses up to ε of hours entirely while delivering perfectly the rest of the time. The same data audited per minute or by overall average looks much better, because the bad hours concentrate the violations.
- *Persistently hard realisations*. Only (1) refuses to concede any single hour. A monthly audit by the worst-hour metric will typically reveal one or two catastrophic hours under (2)–(4), but not under (1).

Take-away. All four formulations “deliver 90%,” but each in a different sense. The mismatch between bid formulation and audit metric is largest when the audit applies a stricter metric than the one the bid was designed for: e.g. a (4)-bid audited by the worst-minute or worst-hour metric, or a (3)-bid audited by the per-scenario metric. Aligning the formulation with the audit metric the regulator actually uses is therefore as important as the choice of ε itself.